

GANESH GUPTA

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Experience

Oracle

Jun 2025 – Present

Software Engineer, AI Platform

- Own multi-step agent workflows (LangGraph, CrewAI, AutoGen, OpenAI Agents SDK), building Python agents that pivot across ambiguous natural-language queries.
- Build full-stack platform capabilities (React/TypeScript frontend, Go/Python backend services) with REST APIs consumed across product teams.
- Combine GraphRAG with vector-based RAG for grounded retrieval, working with entity-relationship style data models at scale.
- Own CI/CD pipelines, containerized deployment (Docker, Kubernetes), and Terraform infrastructure as code on AWS.

QureAI (AI Startup)

Mar 2026 – Present

ML Infrastructure Engineer

- Improved semantic retrieval recall 61% to 89% on a 2K-query eval set by rebuilding the chunking pipeline with hybrid dense-sparse retrieval and re-ranking.
- Built custom evaluators tracking drift, latency, and regression across agent/model versions.

Nomura Research Institute Financial Technologies

Sep 2019 – Aug 2023

Software Engineer

- Architected and operated 25+ production Java Spring Boot microservices exposing REST APIs, serving 1M+ users at 99.9% uptime on AWS (EC2, RDS, SQS).
- Drove a redesign of nightly data aggregation, cutting processing from 8 minutes to 90 seconds via indexed views and batch partitioning on multi-terabyte-scale data.
- Mentored engineers through code reviews and design write-ups, driving consistent engineering standards.

Projects

Distributed KV Store & VPC Control Plane Simulator | github.com/ganeshhgupta/distributed-kv-store

- Built a distributed key-value store in Go/C++ implementing Raft consensus, linearizable reads, and log compaction for fault-tolerant data systems.

LangFetch | tinyurl.com/LangFetch

- LangGraph agent with MCP-compatible tool-use compiling natural-language questions into dependency-ordered SQL subqueries with vector retrieval and hybrid re-ranking; deployed as a reusable MCP server.

RootCause AI | github.com/ganeshhgupta/rootcause-ai

- Multi-agent system diagnosing production incidents via tool-use chains; routes alerts to specialized sub-agents and synthesizes evidence-backed root cause reports.

Education

University of Texas at Arlington

Aug 2023 – May 2025

MS, Computer Science, GPA 4.0/4.0

MCKV Institute of Engineering

Jul 2016 – Jul 2020

BTech, Computer Science

Skills

Languages: Go, Python, TypeScript, Java, SQL

Backend & Data: REST APIs, PostgreSQL, Distributed Systems, Raft Consensus

AI & Agents: LangGraph, LLM Integration, RAG

Cloud & DevOps: AWS, Docker, Terraform, CI/CD, React

Certifications

AWS Certified Developer Associate | GCP Professional MLOps Engineer